सहयोगी प्राध्यापक (माहिती तंत्रज्ञान)
महाराष्ट्र अभियांत्रिकी शिक्षक सेवा, गट-अ
उच्च व तंत्र शिक्षण विभाग
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परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११३४)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक या विषयांमध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**Computational Methods:**
	Statistics and Probability : Moments, Skewness and Kurtosis, Probability, conditional probability, various theoretical distributions like binomial, normal, log-normal, Poisson, gamma distribution, Pearson type I/II distribution test of significance, Gumbel distribution, testing of hypotheses – Large sample tests for mean and proportion, Chi-square test, errors, types of errors. **Regression and Correlation :** Regression and correlation – rank correlation – multiple and partial correlation – analysis of variance-one way and two way classifications – experimental design – Latin square design. **Transforms**: Laplace Transformer: LT of standard function, inversions and their application in Computer Engineering. Fourier Transformer: Fourier integral, Fourier transform and their application in Computer Engineering
2	**Data Structure and Algorithms:**
	Arrays, linked lists, stacks, queues, binary trees, hashing, graphs, sorting & searching techniques, Properties of sparse matrices, Linked list representation of sparse, matrices, Properties of threaded trees, insertion, deletion and traversal, Properties of AVL, trees, rotations, insertion and deletion, Properties of red-black trees, rotations, insertion and deletion, Definition of B- trees, basic operations on B-trees, deleting a key from a B-tree, Properties of Min-max heaps, building a heap, basic operations on heaps, application of min-max heaps.
3	**Database Systems:**
	Concurrency & Recovery Management in Centralized DBMS, Concept of Transaction and its properties, Scheduling of transactions, Conflict operations, Two Phase Locking protocol,Recovery management in Centralized DBMS, Distributed DBMS - Concepts and design, functions and architecture of a DDBMS, distributed relational database design,

	Transparencies in a DDBMS, Date's twelve rules for a DDBMS, Distributed transaction management, distributed concurrency control, distributed deadlock management, distributed database recovery, Replication servers, and Distributed query optimization, Mobile databases.
4	**Information Retrieval:**
	Taxonomy of Information Retrieval models, Classic Information Retrieval, **Alternate set:** Theoretical model, Alternative Algebraic models, Alternative Probabilistic models structured text Retrieval models, Models for browsing. Query and Operations in Information Retrieval: Query structures, Keyboard based querying, Pattern matching, structured queries User relevance feedback, Automatic local analysis, Automatic global analysis. **Indexing and Scoring in Information Systems**: Introduction, Inverted Files, Other Indices for Text, Boolean queries and Introduction to Sequential searching, Scoring, term weighting and the vector space model, Parametric and zone indexes, Weighted zone scoring, Learning weights, The optimal weight, Term frequency and weighting, Inverse document frequency, Tf-idf weighting. The vector space model for scoring, Queries as vectors, computingVector scores, efficient scoring and ranking, In exact top K document retrieval**. Applications of Information Retrieval Systems:** Multimedia Information Retrieval, Distributed Information Retrieval.
5	**Computer Architecture**
	Fundamentals of computer design, instruction set principles and examples, pipelining, advanced pipelining and instruction-level parallelism, memory-hierarchy design and survey of design issues in storage, interconnection network and multiprocessor systems. Quantitative Measure of Performance for Evaluation of Designs, Instruction Set Architecture: Principles and Examples, Process Design: Data path and control units, Pipelining: Advanced design technologies and hazards, Instruction-level parallelism, Memory subsystems: Caches,Input/Output subsystems: Interfacing I/O to CPU/OS, Memory subsystems: Caches, Architectural support for security.
6	**Software Project Management**
	Characteristics of a software project, Software scope and feasibility, resources, the SPM plan,Size/scope estimation, Decomposition techniques, WBS, Sizing, Function point, LOC, FP vs LOC,GANTT Charts, Activity networks, PERT/CPM networks, COCOMO I, COCOMO II models, Quality control, Quality assurance, Formal Technical Reviews, The SQA Plan, ISO and CMM standards,Reactive vs. proactive Risk strategies, Risk projection, Risk Refinement, Risk Monitoring, Monitoring and management, RMMM plan, Earned Value Analysis, Team structures:hierarchical, Egoless, chief programmer, mixed; Team software Process; Resource leveling, Building a team: Skill sets, Configuration Management: Baselines, Configurable items, SCM repository, SCM process, version control change control, configuration audit, Project Monitoring and Control - Audits and Reviews.
7	**Computer and Network Security:**
	Introduction to computer and Network Security, Security policies and security standards, Basic cryptography, Crypto-analysis, Digital certificate creation and signing, PKI, Use of digital certificates in secure email and web system mutual authentication, Network Attacks, Penetration Testing, Backtrack tool set, Nessus network scanner, and Metasploit, Buffer Overflow attacks and defense, Network Defenses, Firewalls, IP tables, IDS, Snort, DDoS and Autonomous Anti-DDos Defense, Access Control, Multilevel Security, SELinux, Privilege Management.
8	**Soft Computing Techniques:**
	Neural Networks History, overview of biological Neuro-system, Mathematical Models of Neurons, ANN architecture, Learning rules, Learning Paradigms-Supervised, Unsupervised and reinforcement Learning, ANN training Algorithms-perceptions, Training rules, Delta, Back Propagation Algorithm, Multilayer Perceptron Model, Hopfield Networks, Associative Memories, Applications of Artificial Neural Networks. Introduction to Fuzzy Logic, Classical and Fuzzy Sets: Overview of Classical Sets, Membership Function, Fuzzy rule generation, Fuzzy Sets: Compliment, Intersections, Unions, Combinations of Operations, Aggregation Operations, Fuzzy Arithmetic, Fuzzy

	Numbers, Linguistic Variables, Arithmetic Operations on Intervals & Numbers, Lattice of Fuzzy Numbers, Fuzzy Equations, Introduction of Neuro-Fuzzy Systems, Architecture of Neuro Fuzzy Networks, Application of Fuzzy Logic: Medicine, Economics etc, An Overview of Genetic Algorithms (GAs), GA operators, GA in problem solving, Implementation of GA.
9	**Artificial Intelligence**
	A.I. Representation, Non-AI & AI Techniques, Representation of Knowledge, Uninformed Search Methods: Depth First Search, Breadth First Search, Depth Limited Search, Iterative Deepening Depth First Search, Bidirectional Search, Comparison of Uninformed search Strategies.Informed Search Methods: Generate & test, Hill Climbing, Best First Search, A* and AO* Algorithm, Constraint satisfaction, Means Ends Analysis, Game playing: Minimax Search, Alpha-Beta Cut offs, Waiting for Quiescence, Logical Agents: Knowledge based agents, Wumpus world. Propositional Logic: Representation, Inference, Reasoning Patterns, Resolution, Forward and Backward Chaining. First order Logic: Representation, Inference, Reasoning Patterns, Resolution, Forward and Backward Chaining. Knowledge Base Systems, State Space Search, Production Systems, Problem Characteristics, Types of production systems, Turing Test. Intelligent Agents: Agents and Environments, concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation. Formulation of problems: Vacuum world, 8 queens, Route finding, robot navigation.
10	**Augment Reality & Virtual Reality**
	Defining **Virtual Reality**, History of VR, Human Physiology and Perception, Key Elements of Virtual Reality Experience, Virtual Reality System, Interface to the Virtual World-Input & output- Visual, Aural & Haptic Displays, Applications of Virtual Reality.Representation of the Virtual World, Visual Representation in VR, Aural Representation in VR and Haptic Representation in VR.The Geometry of Virtual Worlds &The Physiology of Human Vision: Geometric Models, Changing Position and Orientation, Axis-Angle Representations of Rotation, Viewing Transformations, Chaining the Transformations, Human Eye, eye movements & implications for VR.Visual Perception & Rendering: Visual Perception - Perception of Depth, Perception of Motion, Perception of Color, Combining Sources of Information Visual Rendering -Ray Tracing and Shading Models, Rasterization, Correcting Optical Distortions, Improving Latency and Frame Rates. **Augmented Reality**, history of augmented reality, The Relationship Between Augmented Reality and Other Technologies-Media, Technologies, Other Ideas Related to the Spectrum Between Real and Virtual Worlds, applications of augmented reality Augmented Reality Concepts- How Does Augmented Reality Work? Concepts Related to Augmented Reality, Ingredients of an Augmented Reality Experience. Augmented Reality Hardware - Hardware Displays – Audio Displays, Haptic Displays, Visual Displays, Other sensory displays, Visual Perception, Requirements and Characteristics, Spatial Display Model. Processors – Role of Processors, Processor System Architecture, Processor Specifications. Tracking & Sensors - Tracking, Calibration, and Registration, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors, Optical Tracking, Sensor Fusion. Computer Vision for Augmented Reality - Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Simultaneous Localization and Mapping, Outdoor Tracking Augmented Reality Software - Introduction, Major Software Components for Augmented Reality Systems, Software used to Create Content for the Augmented Reality Application.
11	**Embedded System and Internet of Thing**
	Embedded Systems: Architecture & Characteristics of ES, Types of Embedded systems, Examples of Embedded Systems. Embedded System On Chip (SOC). Components of ES: Hardware and software Hardware components of ES: Power supply: types, characteristics, selection criteria, Procesing Unit, Input devices, Output Devices Embedded operating Systems: Operating Systems Concepts, Real time operating systems, and, Task Scheduling, Different OS tasks, Introduction to Real-Time Operating Systems, characteristics, selection criteria, bootloader: U-boot, Sensors: Roles of Sensors & Actuators, Types of sensors, Active and passive, analog and digital,Contact and no-contact, Absolute and relative Working of

	Sensors: Position, occupancy and motion, velocity and acceleration, force, pressure, flow, Acoustic, Humidity, light, radiation, temperature, chemical, biosensor, camera. Development boards: Types of boards - Arduino, Raspberry pi, Beagle bone, ESP8266, selection criteria. Interfacing of sensors with development boards. **Internet of Things:** Definition and characteristics of IoT, Technical Building blocks of IoT, Device,Communication Technologies, Data, Physical design of IoT, IoT enabling technologies, IoT Issues and Challenges- Planning, Costs and Quality, Security and Privacy, Risks, Communication under IoT: IoT Protocols: MQTT, CoAP, XMPP and AMQT, IoT communication models, IoT Communication technologies: Bluetooth, BLE, Zigbee, Zwave, NFC, RFID, LiFi, Wi-Fi, Interfacing of wifi, RFID, Zigbee, NFC with development board. Case Studies: Smart Home, Smart Agriculture, e-health, Smart cities, etc.
12	**Data Science**
	Defining data science and big data, Recognizing the different types of data, Gaining insight into the data science process, Data Science Process: Overview, Different steps, Machine Learning Definition and Relation with Data Science. Statistics: Describing a Single Set of Data, Correlation, Simpson's Paradox, Some Other Correlational Caveats, Correlation and Causation, Probability : Dependence and Independence, Conditional Probability, Bayes's Theorem, RandomVariables, Continuous Distributions, The Normal Distribution, The Central Limit Theorem, Data Analysis Theory and Methods: Clustering –Overview, K-means- overview of method,determining number of clusters, Association Rules- Overview of method, Apriori algorithm,evaluation of association rules, Regression-Overview of linear regression method, model description.Classification- Overview, Naïve Bayes classifier Decision Trees: What Is a Decision Tree? Entropy, The Entropy of a Partition, Creating a Decision Tree, Random Forests Neural Networks: Perceptron, Feed-Forward Neural Networks, Backpropagation, Example: Defeating a CAPTCHA, MapReduce: Why MapReduce? Examples like word count and matrix multiplication, Introduction to data visualization, challenges of data visualization, Definition of Dashboard, Their type, Evolution of dashboard, dashboard design and principles, display media for dashboard. Types of Data visualization: Basic charts scatter plots, Histogram, advanced visualization Techniques like streamline and statistical measures, Plots, Graphs, Networks, Hierarchies, Reports.Data visualization of multidimensional data: Need of data modeling, Multidimensional data models, Mapping of high dimensional data into suitable visualization method- Principal component analysis, clustering study of High dimensional data.
13	**Machine Learning:**
	Supervised Learning -Learning a Class from Examples, Vapnik-Chervonenkis (VC) Dimension, Probably Approximately Correct (PAC) Learning, Noise, Learning Multiple Classes, Regression, Model Selection and Generalization, Dimensions of a Supervised Machine Learning Algorithm, Dimensionality Reduction- Introduction, Subset Selection, Principal Components Analysis, Factor Analysis, Multidimensional Scaling, Linear Discriminant Analysis, Isomap, Locally Linear Embedding. Unsupervised Learning - Introduction, Association Rules-Market Basket Analysis, The Apriori Algorithm, Unsupervised as Supervised Learning, Generalized Association Rules, Cluster Analysis-Proximity Matrices, Clustering Algorithms-K-mean, Gaussian Mixtures as Soft K-means Clustering, Example : Human Tumor Microarray Data, Vector Quantization, K-medoids, Hierarchical Clustering, Self-Organizing Maps, PCA-Spectral Clustering Linear Methods for Regression: Introduction, Linear Regression Models and Least Squares, Subset Selection, Shrinkage Methods-Ridge Regression, Lasso Regression, Least Angle Regression, Methods Using Derived Input Directions-Principal Components Regression, Partial Least Squares, A Comparison of the Selection and Shrinkage Methods , Multiple Outcome Shrinkage and Selection, More on the Lasso and Related Path Algorithms, Logistic Regression-Fitting Logistic Regression Models, Quadratic Approximations and Inference, L1 Regularized Logistic Regression. Support Vector Machines and Tree-Based Models, SVM-Introduction to SVM, The Support Vector Classifier, Support Vector Machines and Kernels-Computing the SVM for Classification, The SVM as a Penalization Method, Function Estimation and Reproducing Kernels, SVMs and the Curse of Dimensionality, A Path

	Algorithm for the SVM Classifier, Support Vector Machines for Regression, Regression and Kernels Tree Based Methods-Regression Trees, Classification Trees, Random Forests- Definition of Random Forests, Details of Random Forests- Out of Bag Samples, Variable Importance, Proximity Plots, Random Forests and Overfitting, Analysis of Random Forests-Variance and the De-Correlation Effect, Bias, Adaptive Nearest Neighbors.
14	**Blockchain Technology**
	What is Blockchain? How does a Blockchain work, The origins of blockchain, Blockchain Applications. Basic Cryptography: Blockchain came from Bitcoin, Why is Blockchain a Distributed, P2P Network?, Blockchain Vs Cryptocurrency, Types of Blockchain. Cryptocurrency and Bitcoin: Different Blockchain Technologies, Benefits of using Blockchain Technology, Blockhain Technology, The Evolution of Blockchain Technology, Blockchain Technology – Basics, Introduction to the Decentralized Web Introduction to Distributed Ledgers, Merkle Tree and Hashing, Blocks, Wallets, and Addresses, Public and Private Key. Consensus Algorithms: Cryptography and Cryptographic Algorithms, Transaction Execution and Distribution, Components of Blockchain Ecosystem, Blockchain Architecture.

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग